

$$\forall n \in \mathbb{Z}^+, \exists x, y, z, w \in \mathbb{Z}, \text{ s.t. } n = x^2 + y^2 + z^2 + w^2.$$

1st: p is odd prime, $\exists a, b \in \mathbb{Z}$ s.t. $a^2 + b^2 + 1 \equiv 0 \pmod{p}$

Proof: Set $A = \{a^2 \mid 0 \leq a \leq \frac{p-1}{2}\}$, $B = \{-b^2 - 1 \mid 0 \leq b \leq \frac{p-1}{2}\}$.

1st if $a_1^2, a_2^2 \in A$, $a_1 \neq a_2$, $a_1^2 \equiv a_2^2 \pmod{p} \Rightarrow \exists k \in \mathbb{Z}$ s.t. $a_1^2 = a_2^2 + kp$

Since $0 \leq a^2 \leq \left(\frac{p-1}{2}\right)^2 \leq \frac{p^2}{4} < p^2$, ~~$\Rightarrow a_1^2 + kp < p^2$~~

if $a_1^2 + kp \leq p^2$, $k > 0 \Rightarrow p^2 - kp - a_1^2 > 0 \Rightarrow k^2 + 4a_1^2 < 0$

which is impossible.

2nd Set $\exists k \in \mathbb{Z}$ s.t. $-b_1^2 - 1 = -b_2^2 - 1 - kp \Rightarrow b_1^2 = b_2^2 + kp$,
same as before.

Since $|A| + |B| \leq \underbrace{\left(\frac{p-1}{2} + 1\right)}_{0 \text{ matters}} \cdot 2 = p+1$, but we can have

at most p distinct numbers to be not mod $p \Rightarrow$

$\exists a^2 \in A, b^2 \in B$, s.t. $a^2 \equiv -b^2 - 1 \pmod{p}$ 12

By Euler's formula,

$$(a^2+b^2+c^2+d^2)(w^2+x^2+y^2+z^2) = (aw+bx+cy+dz)^2 +$$

$$(ax-bw-cz+dy)^2 + (ay+bz-cx-dw)^2 + (az-by+cx-dw)^2$$

Proof: Consider $p = a + bi + cj + dk$
 $q = w + xi + yj + zk$
 LHS: $p\bar{p}q\bar{q}$, RHS: Expansion.

Since $\forall n \in \mathbb{Z}^+$, n can be decomposed into $p_1^{n_1} \dots p_k^{n_k}$, $p_1, \dots, p_k$ are prime numbers. By Euler's formula, if a prime number can be represented by the summation of 4 square numbers, n can be.

Goal: Since $2 = 1^2 + 1^2 + 0^2 + 0^2 \Rightarrow$ Goal changed; $\forall$ odd prime can be represented $= x^2 + y^2 + z^2 + w^2$.

By Lel: $\exists m \in \mathbb{Z}$ s.t. $mp = a^2 + b^2 + c^2 + d^2$

If $m=1$, we are done.

If $m > 1$, w.t. $\exists 0 < m' < m$ s.t. $m'p = a^2 + b^2 + c^2 + d^2$.
 Then by continuously searching such m , we get $m=1$.

If m is even, ① all a, b, c, d even. By $\frac{x^2+y^2}{2} = \left(\frac{x+y}{2}\right)^2 + \left(\frac{x-y}{2}\right)^2$
 ② 2 are even, 2 are odd. Even if a, d is odd, $c^2 + d^2$ is even.
 If so for $mp = a^2 + b^2 + c^2 + d^2 = 2 \left(\frac{a^2+b^2}{2} \right) + 2 \left(\frac{c^2+d^2}{2} \right)$
 $\Rightarrow \frac{mp}{2} = \left(\frac{a+b}{2} \right)^2 + \left(\frac{a-b}{2} \right)^2 + \left(\frac{c+d}{2} \right)^2 + \left(\frac{c-d}{2} \right)^2$

Choose $m' = \frac{mp}{2} < m$.

If m is odd, use $\begin{pmatrix} a \\ b \\ c \\ d \end{pmatrix} \equiv \begin{pmatrix} w \\ x \\ y \\ z \end{pmatrix} \pmod{m}$ and $w, x, y, z \in [\frac{-m}{2}, \frac{m}{2})$

Then we get $w^2 + x^2 + y^2 + z^2 < m^2$ and set $x^2 + w^2 + y^2 + z^2 = \frac{km}{2}$, $k \in \mathbb{Z}$, $0 < k < m$

By Euler's formula, LHS = km^2

Obviously: $m | aw + bx + cy + dz$, $m | \dots$ (rest of RHS)

$$\Rightarrow kp = \sum_i \left(\frac{\text{RHS}_i \text{Term}}{2} \right)^2, \text{ this is } k < m.$$

Then we proved the Lagrange 4-Square Theorem. $\square$

Application: what number can be only represented by 4 square numbers.

Notice if $4n = a^2 + b^2 + c^2 \Rightarrow a, b, c$ must all be even, $n = (\frac{a}{2})^2 + (\frac{b}{2})^2 + (\frac{c}{2})^2$

For $n = 4^k (8m+7)$, if $n = a^2 + b^2 + c^2 \Rightarrow \exists \tilde{a}, \tilde{b}, \tilde{c}$ s.t. $8m+7 = \tilde{a}^2 + \tilde{b}^2 + \tilde{c}^2$

$\Rightarrow 7 \equiv \tilde{a}^2 + \tilde{b}^2 + \tilde{c}^2 \pmod{8}$, however, such 3 numbers doesn't exist.

Sufficient/Necessity: If $n = 4^k (8m+7) \Rightarrow$ ~~only be represent~~ Can't be represented by 3-Square number.

Sufficient: Legendre's 3-Square Theorem.